Machine Learning Concepts 1 (CSE3967)

Assignment-1

(Machine Learning Basics)

Hands-On Exercises

Serial Number	Questions	Course Outcome
1	**Given;** data = np.array([[10, -20, 30], [40, -50, 60], [70, 80, -90]]) **Using the package *Numpy* find out;** • Mean of each row. • Mean of each column. • Standard deviation of each row. • Maximum value of each column. • Minimum value of each row.	CO1
2	**Given;** A = np.array([[2, -4], [-6, 8]]) B = np.array([[1, -3], [-5, -7]]) **Find out;** • $A + B$ • $A - B$ • $A \cdot B$ • A^T • $\det(A)$ • A^{-1}	CO1
3	**From a dictionary, create a Pandas DataFrame containing *Products*, *Price*, *Sales* and *Profit* as columns.** Then perform the following operations:	CO1

	• Print the DataFrame. • Print all the column names. • Count the number of rows and columns. • Print the first three rows. • Print only the *Sales* and *Profits* columns. • Calculate the average, maximum, and minimum sales. • Print the product name whose profit is more than 4000. • Sort the DataFrame according to *Profit* in descending order.	
4	**Using the DataFrame created above, draw the following plots:** • Create a scatter plot of *Price vs Sales* and highlight the product having the highest sales using a different marker. • Create a line plot showing *Sales and Profit* on the same graph and customise the markers and line styles. • Create a bar graph of Profit and display the numerical profit value above each bar. • Create a bar chart showing Profit as a percentage of Sales for each product. **Hint:-** $\text{Profit_margin} = (\text{Profit} * \text{Sales}^{-1}) \times 100$ • Create a pie chart showing the percentage contribution of each product to total sales, with product names as labels. • Create a histogram of *Sales* and add appropriate x-axis and y-axis labels.	CO1

5	**Write a Python program to generate a synthetic dataset containing 100 samples or examples with one feature, namely *Study_Hours* (independent variable), and one target, namely *Test_score* (dependent variable). Follow the instructions given below:** • Generate the dataset using an appropriate random-data generation method, and store the data in a DataFrame. • Print the complete dataset. • Print the following: – Number of samples or examples. – Number of features. – Number of target variables. – Feature name(s). – Target – Shape of the dataset. – Data types of the features. • Print statistics of the dataset, including count, mean, standard deviation, minimum, maximum, and quartiles. • Check whether the dataset contains any missing values. • Create a scatter plot to visualise the relationship between the independent variable and dependent variable. Add an appropriate title, x-axis label, and y-axis label to the visualisation.	CO1
6	**Write a Python program to develop a Simple Linear Regression model using the synthetic dataset generated in question 5, where *Study_Hours* is the independent variable, and *Test_score* is the dependent variable.** **Follow the instructions given below:**	CO2

	• Define the independent variable and dependent variable. Use *Study_Hours* as the feature (X) and *Test_score* as the target (y). • Split the dataset into training and testing sets using an appropriate *train-test* splitting method. Use 80% of the samples for training and 20% for testing. • Print the number of training samples and testing samples. • Create and train a Simple Linear Regression model using the training dataset. • Determine and print the model parameters, including: – Slope (coefficient) – Intercept • Print the mathematical regression equation in the following form: $$Test_score = m \times Study_Hours + c$$ where m represents the slope and c represents the intercept. • Use the trained Linear Regression model to predict the *Test_score* values for the testing dataset. • Evaluate the performance of the regression model using the following evaluation metrics: – Mean Absolute Error (MAE) – Mean Squared Error (MSE) – R^2 Score • Create a Pandas DataFrame containing the actual and predicted values, along with the corresponding *Study_Hours*. Display the complete results.	

	• Save the actual and predicted results to a CSV file named *linear_regression_predictions.csv*. • Create a scatter plot and regression line to visualise the performance of the Simple Linear Regression model. Insert: – An appropriate title. – X-axis label. – Y-axis label. – Legend. – Grid	

Home Assignments

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1	**Given;** MARKS = np.array([[80, 75, 90], [65, 70, 72], [90, 88, 95], [55, 60, 58]]), Where each row represents a student, and each column represents a subject. **Using the package *Numpy* find out;** • Each student's average. • The student with the highest average. • The average for each subject. • The subject with the highest average. • The students whose average is greater than 75.	CO1
2	Using the provided product data, create a comprehensive visualisation showing the relationships among *Price, Sales, and Profit*. Use a scatter plot where *Price* is on the x-axis, Sales is on the y-axis, *Profit* determines the colour and marker size, and Product IDs are annotated on the points.	CO1
3	Write a Python program to load a real-world dataset (in .csv form) from an external source (UCI Machine Learning repositories, Keel dataset repositories, or Kaggle) and store it in a Pandas DataFrame.	CO1

4	**A company has a dataset containing 20000 customer records. Each record has:** • **Age** • **Income** • **Number_of_purchases** • **Membership_duration** • **Website_visits** • **Customer_spending** **The target variable is Customer_spending.** Answer by writing mathematical equations: • Define the total number of samples, N. • Define the number of input variables, p. • Define the input vector $\mathbf{x}_i$ for the i-th sample. • Define the output variable y_i. • Define the input matrix $\mathbf{X}$ and specify its dimensions. • Define the output vector $\mathbf{y}$ and specify its dimensions. • Represent the complete dataset mathematically. • Define the machine learning function that maps the input variables to the output variable. • Write the prediction equation using $\hat{y}_i$. • If 16000 samples are used for training, represent the training dataset mathematically. • Represent the testing dataset mathematically. • Determine the dimensions of $\mathbf{X}_{test}$.	CO2

5	**Write a Python program to develop a Simple Linear Regression model using the `Simple_house_price.csv` dataset. Follow the instructions given below:** • Create a scatter plot to visualise the relationship between the independent variable and dependent variable. Add an appropriate title, x-axis label, and y-axis label to the visualisation. • Define the independent and dependent variables by extracting *Area_sqft* as feature X and *Price* as target y. • Split the dataset into training and testing sets using an appropriate train-test splitting method with 80% training and 20% testing data. Use a fixed *random_state* to make the experiment reproducible. • Print and verify the number of training and testing samples. Additionally, calculate and display the percentage of samples allocated to the training and testing sets. • Create and train a Simple Linear Regression model using the training dataset. After training, calculate and display the training-set predictions and compare the dimensions of the predicted and actual target arrays. • Use the trained Linear Regression model to predict *Price* values for the testing dataset. Create an additional column containing the prediction error (residual) calculated as: $$\text{Residual} = \text{Actual} - \text{Predicted}$$ • Evaluate the regression model using MAE, RMSE and R^2 Score.In addition, determine which of the MAE or RMSE is larger and provide the corresponding numerical values to help identify the influence of larger prediction errors.	CO2